Homework 9: 10/08 Content

MATH 1271, Garrett Credi

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Problems listed as AC are found in the Active Calculus textbook at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as APEX are found in the APEX Calculus Textbook again at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as PAULS-PRACT are found in Paul's Online Math Notes - Review Problems while problems listed as PAULS-ASSIGN are found in Paul's Online Math Notes - Assignment Problems and are indexed via [Topic]-[Section]-[Problem Number]

1 Type 1 - Linearization

Choose 4 of the following problems to solve.

1. AC 1.8.1-4 (each of these counts as one problem)
2. AC 1.8.6-9 (each of these counts as one problem)
3. APEX 2.2.4-6 (each of these counts as one problem)

2 Type 2 - Derivative Rules

Choose 6 of the following problems to solve.

1. AC 2.1.1-11 (each of these count as one problem)

2. APEX 2.3.11-25 EXCEPT 14, 16, 17, 19, and 20 (each of these counts as one problem)

3. PAULS-ASSIGN.Derivatives.Differentiation Formulas.1-41 (each of these counts as one problem)

3 Challenge Problem

Use the limit definition of the derivative to show that the sum/difference rule $\frac{d}{dx}(f(x) \pm g(x)) = \frac{d}{dx}(f(x)) \pm \frac{d}{dx}(g(x))$ and the constant multiple rule $\frac{d}{dx}(c \cdot f(x))$ are true.

Once you've shown that, try and use the limit definition of the derivative to compute $\frac{d}{dx}(f(x) \cdot g(x))$. Hint: It is not simply $\frac{d}{dx}(f(x)) \cdot \frac{d}{dx}(g(x))$!